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JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

KATOL ROAD, NAGPUR

An Autonomous Institute, with NAAC "A" Grade

Basic Science & Humanities Department

2023-24 (EVEN Semester)

Education to Eternity

MID SEM EXAMINATION

Course: B. Tech.

Subject: Engineering Graphics & Design

Subject Code: CSE/DS/AI/2T003

Year/Semester: IInd Semester (First Year)

Max Marks: 20 **Date :- 22/04/2024** **Duration:- 1 Hr**

Instructions to the Students:

1. **All the questions carry marks as indicate.**
2. **Assume suitable data whenever necessary.**
3. **Illustrate your answer whenever necessary with the help of neat sketches.**

CO1. Define various concepts like dimensioning, conventions and standards related to engineering graphics in order to become professionally efficient.
CO2. Interpret drawings of simple machine component in first and third angle of projection systems
CO3. Apply theory of projections in projection of lines, projection of planes and projection of solid.
CO4. Classify solid geometry in different positions.
CO5. Assess the two dimensional and three dimensional drawing in CAD software.
CO6. Create the three dimensional engineering objects into two dimensional drawings and vice versa using CAD software.

SECTION A

1. **All Questions are compulsory.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Enlist the use of center line with neat sketch.	1/CO1	1
Q.2	Draw the CATIA software tool symbol of chamfer and fillet.	2/CO1	1
Q.3	Define hexagon.	1/CO2	1
Q.4	Draw the CATIA software tool symbol of rotate and constraint.	2/CO2	1

SECTION B

1. **All Questions are compulsory.**

Q.5	Draw the projection of point P is 30 mm above H.P and 40 mm in front of VP.	1/CO1	2
Q.6	A 50mm long line AB is parallel to both H.P and V.P. The line is 25mm in front of V.P and 60mm above H.P, draw the projections of the line.	2/CO2	2
Q.7	Show the internal and external angle of nonagon.	1/CO1	2
Q.8	Draw the projection of point S is 35 mm below H.P and 42 mm in front of VP.	2/CO2	2

SECTION C

1. **Solve any two questions.**

Q.9	Construct regular nonagon (9 sides) of 40 mm side by any method.	3/CO3	4
Q.10	A line AB, 70mm long, has its end A 15mm above HP and 20mm in front of VP. It is inclined at 30° to HP and 45°to VP. Draw its projections.	3/CO3	4
Q.11	A Pentagonal plane with a 30mm side has an edge on the HP, the surface of the Plane is inclined at 45° to the HP. Draw it's Projections.	4/CO4	4

(5-2) 3 × ~~180~~ 36 108
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